Hyeong-Geol Shin, Ph.D.

Assistant Professor
Department of Neurology, Johns Hopkins University School of Medicine
F.M. Kirby Research Center, Kennedy Krieger Institute

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ORCID 0000-0001-7876-173X
Google Scholar · Website

Research Interests

Physics-informed Machine Learning | Microstructure Imaging | Quantitative MRI | Diffusion MRI

Academic Appointments

2022 – 2024 **Postdoctoral Fellow**
Department of Radiology, School of Medicine, *Johns Hopkins University*, Baltimore, MD, USA
F.M. Kirby Research Center for Functional Brain Imaging, *Kennedy Krieger Institute*, Baltimore, MD, USA
Advisor: Prof. Peter C. M. van Zijl

2024 – 2026 **Research Associate**
F.M. Kirby Research Center for Functional Brain Imaging, *Kennedy Krieger Institute*, Baltimore, MD, USA

2025 – 2026 **Research Associate**
Department of Neurology and Department of Biomedical Engineering, *Johns Hopkins University*, Baltimore, MD, USA

2026 – present **Assistant Professor**
Department of Neurology, School of Medicine, *Johns Hopkins University*, Baltimore, MD, USA
F.M. Kirby Research Center for Functional Brain Imaging, *Kennedy Krieger Institute*, Baltimore, MD, USA

Education

2016 **B.S. in Electrical and Computer Engineering**
Seoul National University, Seoul, Republic of Korea
Cum Laude

2022 **Ph.D. in Electrical and Computer Engineering** (Integrated M.S./Ph.D. Program)
Seoul National University, Seoul, Republic of Korea
Dissertation: "Biophysical Model-based Quantitative Microstructure Mapping using Magnetic Resonance Imaging"
Advisor: Prof. Jongho Lee · Distinguished Dissertation Award

Honors and Awards

2018 **Magna Cum Laude Merit Award**, ISMRM Annual Meeting

2018 **Finalist**, Trainee Competition, ISMRM White Matter Study Group

2021 **Magna Cum Laude Merit Award**, ISMRM Annual Meeting

2022 **Distinguished Dissertation Award**, Department of Electrical and Computer Engineering, Seoul National University

2022 **Summa Cum Laude Merit Award**, ISMRM Annual Meeting

2022 **Oral Presentation Award (2nd place)**, ISMRM White Matter Study Group

2022 **Best Trainee Abstract Award (2nd place)**, ISMRM Electro-Magnetic Tissue Properties Study Group

2024 **Educational Grant**, ACTRIMS Forum 2024, Americas Committee for Treatment and Research in Multiple Sclerosis

2024 **Best Young Investigator Poster Presentation Award**, ACTRIMS Forum 2024, West Palm Beach, FL, USA

2024 **Best Trainee Competition Award (2nd place)**, ISMRM Electro-Magnetic Tissue Properties Study Group

2024 **Best Oral Presentation on Magnetic Properties (3rd place)**, 2nd Joint Workshop on MR Phase, Magnetic Susceptibility and Electrical Properties Mapping, Santiago, Chile

2024 **Golden Hairball Award for the Most Promising Research**, 28th MRI Division Retreat, Department of Radiology, Johns Hopkins University

2025 **Best Trainee Competition Award (2nd place)**, ISMRM Electro-Magnetic Tissue Properties Study Group

2025 **Oral Presentation Award (1st place)**, ISMRM White Matter Study Group

2026 **Magna Cum Laude Merit Award**, ISMRM Annual Meeting

2026 **Invited Speaker**, Gordon Research Conference on In Vivo Magnetic Resonance

Research Funding

Active

2023 – 2028 **NIH/NIAID R01 AI178774** (PI: C. L. Marvel)
Neuroimaging and blood markers in post treatment Lyme disease with persistent neurologic symptoms
Role: Co-Investigator · 09/20/2023 – 08/31/2028
Total budget: **$2,679,779**

2024 – 2027 **Johns Hopkins Fisher Center Impact Award** (PI: C. L. Marvel)
A Pilot Study of the Overlapping Neurological Effects of Long COVID and Post-Treatment Lyme Disease using Brain Imaging and Blood Markers (019MAR2024-I)
Role: Co-Investigator · 06/30/2024 – 06/29/2027
Total budget: **$250,000**

2024 – 2029 **NIH/NINDS R01 NS082347** (PIs: S. Saidha, P. A. Calabresi)
Imaging Neurodegeneration in Multiple Sclerosis
Role: Co-Investigator · 09/01/2024 – 06/30/2029
Total budget: **$3,145,400**

2026 – 2031 **NIH/NIBIB P41 EB031771** (PIs: P. C. M. van Zijl, H. Lu)
MRI Resource for Physiologic, Metabolic and Anatomic Biomarkers
Role: Co-Investigator · 07/01/2026 – 06/30/2031
Total budget: **$8,597,185**

2026 – 2029 **National Multiple Sclerosis Society Research Grant** (PIs: S. Saidha, H. Shin)
Determining imaging markers of remyelination potential in people with multiple sclerosis using advanced multi-modal magnetic resonance imaging (RG-2601-46024)
Role: **Co-PI** · 10/01/2026 – 09/30/2029
Total budget: **$596,086**

Completed

2017 – 2022 **Global Ph.D. Fellowship**, National Research Foundation of Korea
Quantitative MRI for Future Diagnosis: From MRI to In-vivo Histology
Role: **PI** · 03/01/2017 – 02/28/2022 · Government-funded research fellowship for outstanding Ph.D. candidates
Total budget: **KRW 150,000,000**

Teaching

Spring 2016 **Guest Lecturer and Teaching Assistant**, Signals and Systems (430.306)
Department of Electrical Computer Engineering, Seoul National University

Fall 2019 **Guest Lecturer and Teaching Assistant**, Advanced Bioelectrical and Computer Engineering (M2608.000300)
Department of Electrical Computer Engineering, Seoul National University

2024 – 2025 **Certificate of Completion, Signature Program**, Johns Hopkins Teaching Academy
Center for Teaching Excellence and Innovation, Johns Hopkins University

Fall 2025 **Guest Lecturer**, Plenary Lectures by Distinguished International Scholars
Department of Biomedical Engineering, Yonsei University

Professional Service

Leadership and Organizing

2025 – 2026 **Trainee Representative**, ISMRM Electro-Magnetic Tissue Properties Study Group

2026 **Co-organizer**, Trainee Day of the ISMRM Electro-Magnetic Tissue Properties Study Group (online), April 28, 2026

2026 **Local Organizing Committee**, 3rd Joint Workshop on MR Phase, Magnetic Susceptibility and Electrical Properties Mapping, Annapolis, MD, USA, September 20–23, 2026

2026 – 2027 **Past Trainee Representative**, ISMRM Electro-Magnetic Tissue Properties Study Group

Conference Review

2022, 2024 Joint Workshop on MR Phase, Magnetic Susceptibility and Electrical Properties Mapping

2024 – 2026 ISMRM Annual Meeting

Journal Review

European Journal of Neurology
Imaging Neuroscience
Journal of Magnetic Resonance Imaging
Journal of Neuroimaging
Journal of Neuroscience Methods
Magnetic Resonance in Medicine
Medical Image Analysis
Nature Mental Health
NeuroImage
NeuroImage: Clinical
NMR in Biomedicine
Quantitative Imaging in Medicine and Surgery

Professional Training

2019 **Siemens Healthineers MR IDEA Sequence Programming Course** (MR8SDE)

2022 **Philips MR Clinical Science Workshop Program** for Sequence Development

2022 **Philips MR Clinical Science Workshop Program** for Pulse Programming

Technical Skills

Programming	MATLAB, Python, C++, Linux
Neuroimaging	FSL, ANTs, SPM
MRI	Siemens and Philips pulse sequence programming; MRI physics

Publications

H. Shin in bold · * corresponding author · † co-first author

Peer-Reviewed Papers *journal articles and full-length conference papers*

1. An R_2* model of white matter for fiber orientation and myelin concentration
   J. Lee, **H. Shin**, W. Jung, Y. Nam, S. Oh, J. Lee*
   NeuroImage, 162:269–275, 2017 · doi:10.1016/j.neuroimage.2017.08.050
2. Whole brain g-ratio mapping using myelin water imaging (MWI) and neurite orientation dispersion and density imaging (NODDI)
   W. Jung, J. Lee, **H. Shin**, Y. Nam, H. Zhang, S. Oh, J. Lee*
   NeuroImage, 182:379–388, 2018 · doi:10.1016/j.neuroimage.2017.09.053
3. Differential involvement of nigral subregions in idiopathic Parkinson's disease
   Y. Sung, J. Lee, Y. Nam, **H. Shin**, Y. Noh, D. Shin, E. Kim*
   Human Brain Mapping, 39(1):542–553, 2018 · doi:10.1002/hbm.23863
4. Diagnosis of Early-Stage Idiopathic Parkinson's Disease Using High-Resolution Quantitative Susceptibility Mapping Combined with Histogram Analysis in the Substantia Nigra at 3 T
   E. Kim, Y. Sung, **H. Shin**, Y. Noh, Y. Nam*, J. Lee*
   Journal of Clinical Neurology, 14(1):90–97, 2018 · doi:10.3988/jcn.2018.14.1.90
5. Advances in gradient echo myelin water imaging at 3T and 7T
   H. Shin, S. Oh, M. Fukunaga, Y. Nam, D. Lee, W. Jung, M. Jo, S. Ji, J. Choi, J. Lee*
   NeuroImage, 188:835–844, 2019 · doi:10.1016/j.neuroimage.2018.11.040
6. Initial diagnostic workup of parkinsonism: Dopamine transporter positron emission tomography versus susceptibility map-weighted imaging at 3T
   Y. Sung, J. Lee, Y. Nam, **H. Shin**, Y. Noh, K. Hwang, H. Lee, E. Kim*
   Parkinsonism & Related Disorders, 62:171–178, 2019 · doi:10.1016/j.parkreldis.2018.12.019
7. Artificial neural network for myelin water imaging
   J. Lee, D. Lee, J. Choi, D. Shin, **H. Shin**, J. Lee*
   Magnetic Resonance in Medicine, 83(5):1875–1883, 2020 · doi:10.1002/mrm.28038
8. A geometric approach to separate the effects of magnetic susceptibility and chemical shift/exchange in a phantom with isotropic magnetic susceptibility
   H. Eun, H. Jeong, J. Lee, **H. Shin**, J. Lee*
   Magnetic Resonance in Medicine, 85(1):281–289, 2021 · doi:10.1002/mrm.28408
9. DeepResp: Deep learning solution for respiration-induced B_0 fluctuation artifacts in multi-slice GRE
   H. An, **H. Shin**, S. Ji, W. Jung, S. Oh, D. Shin, J. Park, J. Lee*
   NeuroImage, 224:117432, 2021 · doi:10.1016/j.neuroimage.2020.117432

10. So You Want to Image Myelin Using MRI: An Overview and Practical Guide for Myelin Water Imaging
J. Lee, J. Hyun, J. Lee, E. Choi, **H. Shin**, K. Min, Y. Nam, H. Kim, S. Oh*
Journal of Magnetic Resonance Imaging, 53(2):360–373, 2021 · doi:10.1002/jmri.27059

11. chi-separation: Magnetic susceptibility source separation toward iron and myelin mapping in the brain
H. Shin, J. Lee, Y. Yun, S. Yoo, J. Jang, S. Oh, Y. Nam, S. Jung, S. Kim, M. Fukunaga, W. Kim, H. Choi, J. Lee*
NeuroImage, 240:118371, 2021 · doi:10.1016/j.neuroimage.2021.118371

12. Quad-Contrast Imaging: Simultaneous Acquisition of Four Contrast-Weighted Images (PD-Weighted, T_2-Weighted, PD-FLAIR and T_2-FLAIR Images) With Synthetic T_1-Weighted Image, T_1- and T_2-Maps
S. Ji, J. Jeong, S. Oh, Y. Nam, S. H. Choi, **H. Shin**, D. Shin, W. Jung, J. Lee*
IEEE Transactions on Medical Imaging, 40(12):3617–3626, 2021 · doi:10.1109/TMI.2021.3093617

13. Blip up-down acquisition for spin- and gradient-echo imaging (BUDA-SAGE) with self-supervised denoising enables efficient T_2, T_2*, para- and dia-magnetic susceptibility mapping
Z. Zhang, J. Cho, L. Wang, C. Liao, **H. Shin**, X. Cao, J. Lee, J. Xu, T. Zhang, H. Ye, K. Setsompop, H. Liu, B. Bilgic*
Magnetic Resonance in Medicine, 88(2):633–650, 2022 · doi:10.1002/mrm.29219

14. chi-Separation Imaging for Diagnosis of Multiple Sclerosis versus Neuromyelitis Optica Spectrum Disorder
W. Kim, **H. Shin**, H. Lee, D. Park, J. Kang, Y. Nam, J. Lee, J. Jang*
Radiology, 307(1):e220941, 2023 · doi:10.1148/radiol.220941

15. Depth-wise profiles of iron and myelin in the cortex and white matter using χ-separation: A preliminary study
S. Lee, **H. Shin**, M. Kim, J. Lee*
NeuroImage, 273:120058, 2023 · doi:10.1016/j.neuroimage.2023.120058

16. WaveSep: A Flexible Wavelet-Based Approach for Source Separation in Susceptibility Imaging
Z. Fang†, **H. Shin**†, P. van Zijl, X. Li, J. Sulam*
Machine Learning in Clinical Neuroimaging (MLCN 2023), Lecture Notes in Computer Science, vol. 14312, pp. 56–66, Springer, 2023 · doi:10.1007/978-3-031-44858-4_6 **[Conference paper]**

17. A human brain atlas of χ-separation for normative iron and myelin distributions
K. Min, B. Sohn, W. J. Kim, C. J. Park, S. Song, D. H. Shin, K. W. Chang, N. Shin, M. Kim, **H. Shin**, P. H. Lee, J. Lee*
NMR in Biomedicine, 37(12):e5226, 2024 · doi:10.1002/nbm.5226
Recognized as one of the top 10% most-read papers in NMR in Biomedicine in 2024.

18. Exploiting gradient-echo frequency evolution: Probing white matter microstructure and extracting bulk susceptibility-induced frequency for quantitative susceptibility mapping
L. Chen, **H. Shin**, P. van Zijl, X. Li*
Magnetic Resonance in Medicine, 91(4):1676–1693, 2024 · doi:10.1002/mrm.29958

19. Compartmental anisotropy of filtered exchange imaging (FEXI) in human white matter: What is happening in FEXI?
H. Shin, X. Li, H. Heo, L. Knutsson, F. Szczepankiewicz, M. Nilsson, P. van Zijl*
Magnetic Resonance in Medicine, 92(2):660–675, 2024 · doi:10.1002/mrm.30086

20. Comparison between R_2'-based and R_2*-based χ-separation methods: A clinical evaluation in individuals with multiple sclerosis
S. Ji, J. Jang, M. Kim, H. Lee, W. Kim, J. Lee, **H. Shin***
NMR in Biomedicine, 37(9):e5167, 2024 · doi:10.1002/nbm.5167
Recognized as one of the top 10 most-cited papers in NMR in Biomedicine in 2024.

21. Quantifying Remyelination Using χ-Separation in White Matter and Cortical Multiple Sclerosis Lesions
J. Müller, P. Lu, A. Cagol, E. Ruberte, **H. Shin**, M. Ocampo-Pineda, X. Chen, C. Tsagkas, M. Barakovic, R. Galbusera, M. Weigel, S. Schädelin, Y. Wang, T. Nguyen, P. Spincemaille, L. Kappos, J. Kuhle, J. Lee, C. Granziera*
Neurology, 103(6):e209604, 2024 · doi:10.1212/WNL.0000000000209604
Highlighted in an accompanying editorial: A. Rovira and D. Pareto, "χ-Separation as a Novel MRI Biomarker for Assessing Disease Progression in Multiple Sclerosis," Neurology, 103(6):e209735, 2024. doi:10.1212/WNL.0000000000209735

22. ProxiMO: Proximal Multi-operator Networks for Quantitative Susceptibility Mapping
S. Orenstein, Z. Fang, **H. Shin**, P. van Zijl, X. Li, J. Sulam*
Machine Learning in Clinical Neuroimaging (MLCN 2024), Lecture Notes in Computer Science, vol. 15266, pp. 13–23, Springer, 2025 · doi:10.1007/978-3-031-78761-4_2 **[Conference paper]**

23. Blood-brain barrier breakdown in brain ischemia: Insights from MRI perfusion imaging
S. Sasannia*, R. Leigh, P. B. Bastani, **H. Shin**, P. van Zijl, L. Knutsson, P. Nyquist
Neurotherapeutics, 22(1):e00516, 2025 · doi:10.1016/j.neurot.2024.e00516

24. Association of iron deposition in MS lesion with remyelination capacity using susceptibility source separation MRI
H. Shin, W. Kim, J. Lee, H. Lee, Y. Nam, J. Kim, X. Li, P. van Zijl, P. Calabresi, J. Lee, J. Jang*
NeuroImage: Clinical, 45:103748, 2025 · doi:10.1016/j.nicl.2025.103748

25. chi-sepnet: Deep Neural Network for Magnetic Susceptibility Source Separation
M. Kim, S. Ji, J. Kim, K. Min, H. Jeong, J. Youn, T. Kim, J. Jang, B. Bilgic, **H. Shin***, J. Lee*
Human Brain Mapping, 46(2):e70136, 2025 · doi:10.1002/hbm.70136

26. In-vivo high-resolution χ-separation at 7T
J. Kim, M. Kim, S. Ji, K. Min, H. Jeong, **H. Shin**, C. Oh, R. J. Fox, K. E. Sakaie, M. J. Lowe, S. Oh, S. Straub, S. Kim, J. Lee*
NeuroImage, 308:121060, 2025 · doi:10.1016/j.neuroimage.2025.121060

27. Resolution generalization of deep learning-based dipole inversion networks for QSM
S. Ji, M. Kim, J. Lee*, **H. Shin***
NeuroImage, 324:121598, 2025 · doi:10.1016/j.neuroimage.2025.121598

28. In-vivo iron mapping in patients with Parkinson's disease using deep learning-based susceptibility source separation MRI
H. Shin, K. A. Mills, T. M. Dawson, T. Kim, J. Lee, X. Li, P. van Zijl, J. Prasuhn*
npj Parkinson's Disease, 12(1):200, 2026 · doi:10.1038/s41531-026-01384-x

29. Threshold Effects, Cognitive Decline, and Longitudinal Changes in White Matter Hyperintensity Volume
S. Sasannia, M. Matsyuk, S. Wang, J. Zhang, K. A. Walker, **H. Shin**, R. Leigh, J. Prince, L. Becker, L. Yanek, N. Armstrong, P. van Zijl, D. Vaidya, L. Knutsson, P. Nyquist*
Stroke, 57(9):2695–2708, 2026 · doi:10.1161/STROKEAHA.125.054810

30. Evidence of Iron Accumulation in Cerebral Adrenoleukodystrophy: A Potential Novel Disease Mechanism
C. L. Nemeth, M. Sisman, J. Zhang, **H. Shin**, X. Li, B. Turk, A. Fatemi, T. Nguyen, E. Mallack*
Annals of Clinical and Translational Neurology, 13(6):1276–1282, 2026 · doi:10.1002/acn3.70346

Manuscripts Under Review

1. Blood-Brain Barrier Dysfunction and Microstructural Injury in the Penumbra Surrounding WMH in cSVD
S. Sasannia†, **H. Shin**†, R. Leigh, M. Matsyuk, S. Wang, J. Zhang, D. Danshvaziri, J. Prince, L. Becker, L. R. Yanek, D. Vaidya, F. Szczepankiewicz, P. van Zijl, L. Knutsson, P. Nyquist*
Under review, *Brain Communications*

2. Microstructural MRI characteristics of white matter lesions are associated with clinical disability and retinal layer thicknesses in people with multiple sclerosis
A. Bacchetti†, **H. Shin†**, J. Zhang, G. Moretti, T. Lin, B. McCormack, E. Kinder, N. Lee, A. Filippatou, M. O'Donnell, C. Lambert, G. Gakis, N. Pellegrini, S. Davis, G. Otero-Duran, A. Gulati, E. Lawrence, A. Didouchevski, M. Inserra, B. Dewey, X. Li, H. Lu, J. Xu, J. Hua, F. Szczepankiewicz, A. Carass, J. Prince, K. Fitzgerald, E. Sotirchos, P. van Zijl, P. Calabresi, S. Saidha*
Under revision, *Multiple Sclerosis Journal*

3. Submillimeter postmortem and in vivo diffusion and susceptibility magnetic resonance imaging to characterize cortical micro- and meso-structures
Y. Uchida, **H. Shin**, L. Gomez-Isaza, J. C. Troncoso, A. Bibic, J. Gerold, D. Sung, Y. Jun, F. Wang, Z. Dong, S. Huang, B. Bilgic, P. van Zijl, X. Li*, K. Oishi*
Under revision, *Scientific Reports* · Preprint: doi:10.21203/rs.3.rs-9612742/v1

4. MWF-MIMOSA for efficient simultaneous relaxometry and myelin water fraction mapping
Y. Chen, Y. Jun*, **H. Shin**, S. Li, S. Fujita, X. Yong, J. Kim, J. Lee, G. F. Piredda, T. Hilbert, A. Bhatt, S. Y. Huang, H. Liu*, H. Ye, S. Nasr, B. Gagoski, K.-S. Chan, B. Bilgic
Under review, *IEEE Transactions on Biomedical Engineering* · Preprint: arXiv:2607.28984

5. Multimolecular and multiparametric imaging in multiple sclerosis using saturation transfer MR fingerprinting
S. Mahmud, K. Ju, **H. Shin**, P. A. Calabresi, P. van Zijl, S. Saidha, H. Heo*
Under review, *NMR in Biomedicine*

6. Inter- and intrarater agreement for the identification of paramagnetic rim lesions in people with multiple sclerosis
N. Lee, A. Bacchetti, G. Moretti, M. Martire, **H. Shin**, J. Zhang, M. O'Donnell, C. Lambert, T. Lin, B. McCormack, S. Davis, G. Otero-Duran, N. Pellegrini, A. Didouchevski, A. Filippatou, A. Gulati, P. Calabresi, E. Sotirchos, E. Mowry, K. Fitzgerald, S. Remedios, K. Bartz, J. Prince, A. Carass, L. Zhou, P. van Zijl, M. Absinta, B. Dewey, S. Saidha*
Under review, Multiple Sclerosis Journal

Book Chapter

1. Susceptibility Weighted Imaging and Quantitative Susceptibility Mapping
H. Shin, L. Chen, X. Li
In *Advanced MR Techniques for Neurodegenerative Diseases*, K. W. Y. Chan, P. G. Unschuld, P. C. M. van Zijl, L. Knutsson (Eds.), Elsevier, 2026 · ISBN 978-0-443-21425-7

PATENTS

1. Method for reconstructing QSM MAP using dipole compensation
S. Ji, **H. Shin**, J. Lee (inventors); SNU R&DB Foundation (assignee)
Republic of Korea Patent No. 10-2771508, registered February 18, 2025

2. Method for reconstructing QSM MAP using dipole compensation
S. Ji, **H. Shin**, J. Lee (inventors); SNU R&DB Foundation (assignee)
U.S. Patent No. 12,411,196 B2, issued September 9, 2025 (published as US 2024/0210506 A1)

3. Method for reconstructing paramagnetic and diamagnetic susceptibility maps using multi-compartment modeling
H. Shin, X. Li, P. van Zijl (inventors); Johns Hopkins Technology Ventures (JHTV) (assignee)
U.S. Provisional Patent Application (pending)

SOFTWARE

- SMwI: Susceptibility map-weighted imaging package for nigrosome 1 imaging
list.snu.ac.kr/index.php/software
- chi-separation: Magnetic susceptibility source separation software package
github.com/SNU-LIST/chi-separation

- B_0 navigator correction: Software to compensate for temporal B_0 field fluctuation using navigator echoes
  list.snu.ac.kr/index.php/software

Invited Talks

Jan 28, 2021 "Separating positive and negative susceptibility sources in quantitative susceptibility mapping"
Seoul St. Mary's Hospital, Seoul, Republic of Korea

Aug 23, 2021 "Microstructure imaging using susceptibility source separation"
University of Southern California (online)

Sep 1, 2021 "Exploring brain microstructure using MRI: Myelin water imaging and magnetic susceptibility source separation"
Johns Hopkins University (online)

Sep 7, 2021 "Quantitative microstructural imaging using MRI: Toward iron and myelin mapping"
National Institutes of Health (online)

Apr 18, 2022 "Quantitative microstructural imaging using MRI: Toward iron and myelin mapping"
Johns Hopkins University (online)

May 7, 2022 "T_2/T_2*: Applications"
ISMRM 2022 Weekend Course, Microstructure: Relaxation, Magnetization Transfer & Susceptibility, London, UK

May 12, 2022 "Imaging multiple sclerosis histology using susceptibility source separation: a postmortem brain study"
ISMRM White Matter Study Group, London, UK

May 12, 2022 "Magnetic Susceptibility Imaging of Brain Microstructure Using Chi-Separation"
ISMRM 2022 Member-Initiated Symposium, Bridging the Scales of Tissue Microstructure with Ultra-High Field MRI & Multi-Modal Imaging, London, UK

May 25, 2022 "Brain micro-structure mapping using magnetic susceptibility"
Kennedy Krieger Institute, Baltimore, MD, USA

Jun 3, 2022 "Imaging multiple sclerosis histology using susceptibility source separation: a postmortem brain study"
ISMRM Electro-Magnetic Tissue Properties Study Group (online)

Sep 2, 2022 "Imaging iron and myelin in brain using MRI"
Division of Neuroimmunology and Neurological Infections, Johns Hopkins Hospital, Baltimore, MD, USA

Sep 20, 2022 "Susceptibility tensor imaging using chi-separation"
Johns Hopkins University (online)

Feb 24, 2023 "Microstructure imaging using MRI"
Seoul National University, Seoul, Republic of Korea

Mar 6, 2023 "Susceptibility source separation"
U.S. Food and Drug Administration (online)

Mar 16, 2023 "Chi-Separation: Toward Iron and Myelin Mapping in the Brain"
ISMRM Electro-Magnetic Tissue Properties Study Group Virtual Meeting: Source Separation in QSM

Mar 23, 2023 "Magnetic susceptibility source separation: Toward brain microstructure mapping"
Kennedy Krieger Institute, Baltimore, MD, USA

Oct 30, 2024 "Towards New Imaging Biomarkers for Multiple Sclerosis"
F.M. Kirby Research Center 25th Anniversary Symposium, Kennedy Krieger Institute, Baltimore, MD, USA

Jul 16, 2026 "Beyond QSM: Separating Magnetic Susceptibility Sources"
Gordon Research Conference on In Vivo Magnetic Resonance, Proctor Academy, Andover, NH, USA

Sep 22, 2026 "Update on QSM source separation"
3rd Joint Workshop on MR Phase, Magnetic Susceptibility and Electrical Properties Mapping, Annapolis, MD, USA

Conference Presentations

ISMRM: International Society for Magnetic Resonance in Medicine · Presentation type in italics

2016

1. Simultaneous Generation of Quantitative Susceptibility Map (QSM) and Myelin Water Image (MWI) using Multi-echo GRE Data for Estimation of Iron and Myelin Distribution in Brain
H. Shin, J. Y. Choi, J. Lee. 4th International QSM Workshop, Graz, Austria, 2016. *Poster*

2017

2. Development and Systematic Analysis of 2D and 3D GRE Myelin Water Imaging
H. Shin, S. Oh, J. Lee. 25th ISMRM Annual Meeting, Honolulu, HI, USA, 2017. *Poster*
3. Automation of Susceptibility Map-weighted Imaging (SMwI) for Nigrosome-1 in the Substantia Nigra
H. Shin, E. Kim, Y. Nam, D. Shin, J. Lee. International Congress on Magnetic Resonance Imaging (ICMRI), Seoul, Republic of Korea, 2017. *Oral*
4. Assessment of Patients Who Presented with Parkinsonism: Nigrosome-1 Susceptibility Map-weighted Imaging versus ^{18}F-FP-CIT PET
I. Lee, E. Kim, C. Kim, T. Seong, **H. Shin**, Y. Nam. Radiological Society of North America (RSNA) Annual Meeting, Chicago, IL, USA, 2017. *Poster*
5. Separating positive and negative susceptibility sources in QSM
J. Lee, Y. Nam, J. Choi, **H. Shin**, T. Hwang, J. Lee. 25th ISMRM Annual Meeting, Honolulu, HI, USA, 2017. *Oral*

2018

6. A Sequence for High Quality Gradient Echo Myelin Water Imaging (GRE-MWI) at 3T and 7T
H. Shin, S. Oh, M. Fukunaga, D. Lee, Y. Nam, S. Ji, W. Jung, J. Lee. 26th ISMRM Annual Meeting, Paris, France, 2018. *Oral* · ***Magna Cum Laude Merit Award***
7. A Sequence for High Quality Gradient Echo Myelin Water Imaging (GRE-MWI) at 3T and 7T
H. Shin, S. Oh, M. Fukunaga, D. Lee, Y. Nam, S. Ji, W. Jung, J. Lee. ISMRM Workshop on Advanced Neuro MR: Best Practices for Technical Implementation, Seoul, Republic of Korea, 2018. *Poster*
8. Systematic development of a sequence for gradient-echo-based myelin water imaging (GRE-MWI)
H. Shin, S. Oh, K. Min, J. Lee. International Congress on Magnetic Resonance Imaging (ICMRI), Seoul, Republic of Korea, 2018. *Oral*
9. Analysis of Knee Cartilage using Magnetization Transfer and Multi-exponential T_2* Fitting
S. Ji, S. Oh, Y. Lee, D. Shin, D. Lee, T. Hwang, W. Jung, **H. Shin**, J. Lee. 26th ISMRM Annual Meeting, Paris, France, 2018. *Poster*
10. SafeNet: Artificial Neural Network for Real-Time T_2 Mapping with Quality Assurance
D. Lee, W. Jung, J. Lee, J. Ko, **H. Shin**, H. Eun, Y. Nam, J. Lee. 26th ISMRM Annual Meeting, Paris, France, 2018. *Oral*
11. A procedure to scale myelin water fraction to myelin volume fraction for g-ratio mapping
W. Jung, **H. Shin**, J. Lee, S. Ji, D. Lee, Y. Nam, J. Lee. 26th ISMRM Annual Meeting, Paris, France, 2018. *Poster*

12. Magnetic susceptibility source separation using multi-echo GRE data only
T. Hwang, J. Lee, **H. Shin**, D. Lee, J. Choi, H. Eun, Y. Nam, J. Lee. 26th ISMRM Annual Meeting, Paris, France, 2018. *Poster*

2019

13. Development and Systematic Analysis of 2D and 3D GRE Myelin Water Imaging
H. Shin, S. Oh, Y. Nam, J. Lee. Gordon Research Conference on Tissue Microstructure Imaging, South Hadley, MA, USA, 2019. *Poster*

14. Separating positive and negative susceptibility sources in QSM
H. Shin, J. Lee, Y. Nam, J. Jang, S. Oh, J. Lee. 5th International QSM Workshop, Seoul, Republic of Korea, 2019. *Poster*

15. A Sequence for High Quality Gradient Echo Myelin Water Imaging (GRE-MWI) at 3 T and 7 T
H. Shin, S. Oh, M. Fukunaga, Y. Nam, J. Lee. 5th International QSM Workshop, Seoul, Republic of Korea, 2019. *Poster*

16. Analysis of Gradient Echo myelin water imaging (GRE-MWI) for water exchange and scan parameters
H. Shin, S. Oh, J. Y. Choi, K. Min, H. Eun, J. Lee. 27th ISMRM Annual Meeting, Montréal, QC, Canada, 2019. *Poster*

17. Validation of magnetic susceptibility source separation: Monte Carlo simulation and phantom experiment
J. Lee, **H. Shin**, H. Eun, D. Shin, J. Lee. 27th ISMRM Annual Meeting, Montréal, QC, Canada, 2019. *Poster*

18. Real-time processing of myelin water imaging using artificial neural network
J. Lee, D. Lee, J. Y. Choi, D. Shin, **H. Shin**, J. Lee. 27th ISMRM Annual Meeting, Montréal, QC, Canada, 2019. *Oral*

2020

19. Multi-seed myelin water imaging using gradient echo: beyond the initial guess in exponential-sum fit
H. Shin, S. Oh, S. Ji, J. Lee, W. Jung, E. Choi, J. Lee. 28th ISMRM Annual Meeting, virtual meeting, 2020. *Poster*

20. Machine designing: a multi-coil local shim design using Bayesian optimization
C. Oh, E. Choi, W. Jung, **H. Shin**, J. Lee. 28th ISMRM Annual Meeting, virtual meeting, 2020. *Poster*

21. Closer look at multiple sclerosis lesions: an initial result of positive and negative magnetic susceptibility separation
J. Jang, **H. Shin**, Y. Nam, J. Lee, J. Lee, W. Kim. 28th ISMRM Annual Meeting, virtual meeting, 2020. *Oral*

2021

22. Magnetic susceptibility source separation: validation with Monte-Carlo simulation & application to human brain with histological comparison
H. Shin, Y. Yun, S. Yoo, S. Jung, S. Kim, H. Choi, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Poster* · ***Magna Cum Laude Merit Award***

23. Validation of magnetic susceptibility source separation: Monte-Carlo simulation and Histological comparison in human brains
H. Shin, Y. Yun, S. Yoo, S. Jung, S. Kim, H. Choi, J. Lee. 3rd Annual Scientific Meeting of the Asian Society for Magnetic Resonance in Medicine (ASMRM), Yokohama, Japan, 2021. *Oral*

24. DeepSlider: Deep learning-powered gSlider for improved robustness and performance
J. Park, D. Shin, **H. Shin**, J. Kim, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Poster*

25. Generalization of deep learning-based QSM by expanding the diversity of spatial gradient in training data
W. Jung, S. Bollmann, S. Oh, **H. Shin**, S. Ji, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Poster*

26. AutoSON: Automated Sequence Optimization by joint training with a Neural network
H. An, D. Shin, **H. Shin**, W. Jung, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Poster*

27. DeepResp: Deep Neural Network for respiration-induced artifact correction in 2D multi-slice GRE
H. An, **H. Shin**, S. Ji, W. Jung, S. Oh, D. Shin, J. Park, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Poster*

28. ChEST: A novel model measuring both Chemical Exchange and Susceptibility Tensor from resonance frequency shift
H. Jeong, **H. Shin**, X. Li, S. Ji, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Oral*

29. DeepTSE-T2: Deep learning-powered T_2 mapping with B_1^+ estimation using a product double-echo Turbo Spin Echo sequence
H. Jeong, **H. Shin**, S. Ji, J. Jang, H. Lee, Y. Nam, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Oral*

30. High neuromelanin contrast achieved using sandwiched flow saturation RF pulses: sandwich-fsNM imaging
S. Ji, E. Choi, E. Kim, D. Shin, **H. Shin**, J. Lee. 29th ISMRM Annual Meeting, virtual meeting, 2021. *Poster*

2022

31. Imaging multiple sclerosis histopathology using susceptibility source separation: a postmortem brain study
H. Shin, R. Galbusera, J. Seo, S. Ji, E. Bahn, J. Franz, C. Stadelmann, P. Lu, J. Jang, Y. Lee, C. Granziera, J. Lee. 30th ISMRM Annual Meeting, London, UK, 2022. *Oral* · ***Summa Cum Laude Merit Award***

32. chi-separation using multi-orientation data in invivo and exvivo brains: Visualization of histology up to the resolution of 350 μm
H. Shin, J. Seo, Y. Lee, H. Jeong, S. Ji, M. Kim, J. Park, B. Kim, K. Lee, S. Kim, J. Jang, M. Woo, J. Lee. 30th ISMRM Annual Meeting, London, UK, 2022. *Oral*

33. QALAS + QSM: Efficient Multi-parameter Mapping Allows Disentangling Para- and Dia-magnetic Contributions in Brain Tissue
H. Shin, J. Chen, J. Lee, T. Kim, J. Cho, G. Varela-Mattatall, B. Gagoski, C. Liu, B. Bilgic. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

34. Imaging ex-vivo cynomolgus monkey brain in 150 μm isotropic resolution
C. Oh, Y. Kim, J. Park, Y. Song, C. Jeon, K. Lim, J. Seo, J. Seong, J. Cho, Y. Lee, J. Lee, **H. Shin***. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

35. Application of susceptibility source separation (chi-separation) to UK Biobank protocol and clinical protocol using deep neural network
H. Jeong, S. Oh, J. Lee, **H. Shin***. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

36. Iron and Myelin Content in Multiple Sclerosis Lesions using Magnetic Susceptibility Source Separation
R. Rahmanzadeh, P. Lu, **H. Shin**, M. Weigel, T. D. Nguyen, Y. Wang, F. La Rosa, M. Bach Cuadra, E. Radue, J. Kuhle, L. Kappos, J. Lee, C. Granziera. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

37. Chi-sepnet: Susceptibility source separation using deep neural network
M. Kim, **H. Shin**, C. Oh, H. Jeong, S. Ji, H. An, J. Kim, J. Jang, B. Bilgic, J. Lee. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

38. Laminar profiling in advanced susceptibility imaging reveals variations in iron and myelin concentrations
S. Lee, **H. Shin**, J. Lee. 30th ISMRM Annual Meeting, London, UK, 2022. *Oral*

39. Hypo-diamagnetism on chi-separation: a potential marker for the differential diagnosis between MS and NMOSD
J. Jang, **H. Shin**, H. Lee, D. Park, J. Kang, Y. Nam, J. Lee, W. Kim. 30th ISMRM Annual Meeting, London, UK, 2022. *Oral*

40. Coil to Coil: Self-supervised denoising using phased-array coil images
J. Park, D. Park, **H. Shin**, E. Choi, D. Shin, S. Y. Chun, J. Lee. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

41. Observing hyperintensity region within substantia nigra in cynomolgus monkey
C. Oh, C. Jeon, J. Seo, **H. Shin**, S. Ji, S. Park, S. Song, D. Shin, Y. Lee, J. Lee. 30th ISMRM Annual Meeting, London, UK, 2022. *Poster*

42. Imaging multiple sclerosis histology using susceptibility source separation: a postmortem brain study
H. Shin, R. Galbusera, J. Seo, S. Ji, E. Bahn, J. Franz, C. Stadelmann, P. Lu, J. Jang, Y. Lee, X. Li, P. van Zijl, C. Granziera, J. Lee. 1st Joint Workshop on MR Phase, Magnetic Susceptibility and Electrical Properties Mapping, Lucca, Italy, 2022. *Oral*

2023

43. Diffusion filters, T_2-filters, diffusion tensor component diffusivities, and compartmental diffusivities. What is happening in FEXI?
H. Shin, X. Li, H. Heo, L. Knutsson, F. Szczepankiewicz, M. Nilsson, P. van Zijl. 31st ISMRM Annual Meeting, Toronto, ON, Canada, 2023. *Poster*

44. A toolbox of chi-separation for magnetic susceptibility source separation
J. Lee, **H. Shin**, M. Kim, J. Lee, S. Oh. 31st ISMRM Annual Meeting, Toronto, ON, Canada, 2023. *Poster*

45. Comparison of chi-separation and chi-separation*: A clinical feasibility study
J. Jang, **H. Shin**, M. Kim, H. Lee, W. Kim. 31st ISMRM Annual Meeting, Toronto, ON, Canada, 2023. *Poster*

46. A feasibility study of susceptibility source separation via chi-separation in amyotrophic lateral sclerosis patients at 7T
J. Kim, **H. Shin**, S. Ji, H. Jeong, H. An, C. Sohn, S. Han, H. Song, J. Chae, S. Choi, J. Sung, J. Lee. 31st ISMRM Annual Meeting, Toronto, ON, Canada, 2023. *Poster*

47. Successful generalization for data with higher or lower resolution than training data resolution in deep learning powered QSM reconstruction
S. Ji, J. Park, **H. Shin**, J. Yoon, M. Kim, J. Lee. 31st ISMRM Annual Meeting, Toronto, ON, Canada, 2023. *Poster*

48. In-vivo delineation of fine structures in the human brain using high resolution deep learning-powered chi-separation
S. Ji, J. Park, **H. Shin**, M. Kim, J. Lee. 31st ISMRM Annual Meeting, Toronto, ON, Canada, 2023. *Oral*

49. Identifying Multiple Sclerosis Lesion Subtypes with Distinct Microstructural Features using Advanced Microstructural MRI and Unsupervised Machine Learning
H. Shin, B. Dewey, J. Brabec, X. Li, O. Ezzedin, K. Ecoff, A. Ramirez, A. DuVal, K. Fitzgerald, J. Prince, S. Saidha, P. Calabresi, P. van Zijl. 9th Joint ECTRIMS-ACTRIMS Meeting, Milan, Italy, 2023. *Poster*

2024

50. Identifying Multiple Sclerosis Lesion Subtypes with Distinct Microstructural Features using Advanced Microstructural MRI
H. Shin, B. E. Dewey, J. Brabec, J. Zhang, O. Ezzedin, K. Ecoff, A. Kim, A. Ramirez, A. DuVal, K. Fitzgerald, L. Knutsson, F. Szczepankiewicz, J. L. Prince, S. Saidha, P. Calabresi, P. van Zijl, X. Li. North American Imaging in Multiple Sclerosis (NAIMS) Cooperative Workshop, West Palm Beach, FL, USA, 2024. *Poster*

51. Identifying Multiple Sclerosis Lesion Subtypes with Distinct Microstructural Features using Advanced Microstructural MRI
H. Shin, B. E. Dewey, J. Brabec, J. Zhang, O. Ezzedin, K. Ecoff, A. Kim, A. Ramirez, A. DuVal, K. Fitzgerald, L. Knutsson, F. Szczepankiewicz, J. L. Prince, S. Saidha, P. Calabresi, P. van Zijl, X. Li. ACTRIMS Forum 2024, West Palm Beach, FL, USA, 2024. *Poster*

52. Association of Iron Deposition in Early-Stage Multiple Sclerosis Lesion with Remyelination Capacity: chi-separation imaging study
H. Shin, W. Kim, H. Lee, J. Kim, Y. Nam, X. Li, P. van Zijl, J. Lee, J. Jang. 32nd ISMRM Annual Meeting, Singapore, 2024. *Oral*

53. Identifying Multiple Sclerosis Lesion Subtypes with Distinct Microstructural Features using Advanced Microstructural MRI
H. Shin, B. E. Dewey, J. Brabec, J. Zhang, O. Ezzedin, K. Ecoff, A. Kim, A. Ramirez, A. DuVal, K. Fitzgerald, L. Knutsson, F. Szczepankiewicz, J. Prince, S. Saidha, P. A. Calabresi, P. van Zijl, X. Li. 32nd ISMRM Annual Meeting, Singapore, 2024. *Oral*

54. Systematic analysis of relaxometric constant in brain using temperature-dependent relaxometry and susceptibility: Toward 7T chi-separation
H. Shin, Y. Uchida, J. Redding-Ochoa, K. Onda, S. Ji, A. Barrett, A. Bibic, J. C. Troncoso, J. Kim, P. van Zijl, K. Oishi, X. Li. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

55. Microscopic diffusion anisotropy as a predictor of cognitive decline in asymptomatic adults
H. Shin, S. Sasannia, S. Mahmud, M. Matsyuk, S. Wang, J. Zhang, F. Szczepankiewicz, X. Li, J. Prince, L. Knutsson, P. van Zijl, P. Nyquist. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

56. Visualization of iron and myelin cytoarchitecture in postmortem human brain using sub-millimeter multi-orientation chi-separation
H. Shin, Y. Uchida, J. Redding-Ochoa, K. Onda, A. Barrett, A. Bibic, J. C. Troncoso, P. van Zijl, K. Oishi, X. Li. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

57. Investigation and validation for cortical laminar structures of myelin and iron using χ-separation (chi-separation)
B. Moon, **H. Shin**, J. Kim, S. Ji, C. Oh, J. Lee. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

58. In-vivo high-resolution χ-separation (chi-separation) at 7T
J. Kim, **H. Shin**, M. Kim, S. Ji, K. Min, H. Jeong, S. Kim, J. Lee. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

59. A human brain atlas of χ-separation (chi-separation) for normative iron and myelin distributions
K. Min, B. Sohn, W. J. Kim, C. J. Park, S. Song, D. H. Shin, K. W. Chang, N. Shin, M. Kim, **H. Shin**, P. H. Lee, J. Lee. 32nd ISMRM Annual Meeting, Singapore, 2024. *Oral*

60. DeepSepSTI: Improved Susceptibility Tensor Reconstruction by Anisotropic Susceptibility Source Separation
Z. Fang, **H. Shin**, B. E. Dewey, P. A. Calabresi, P. van Zijl, J. Sulam, X. Li. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

61. Susceptibility tensor imaging of ex vivo human hemibrain using 7T human MR scanner
Y. Uchida, **H. Shin**, J. Redding-Ochoa, K. Onda, A. Barrett, A. Bibic, J. Troncoso, P. van Zijl, K. Oishi, X. Li. 32nd ISMRM Annual Meeting, Singapore, 2024. *Poster*

62. Systematic analysis of relaxometric constants in brain tissue using temperature-dependent transverse relaxometry and magnetic susceptibility
H. Shin, Y. Uchida, J. Redding-Ochoa, K. Onda, A. Barrett, A. Bibic, J. C. Troncoso, P. van Zijl, K. Oishi, X. Li. 2nd Joint Workshop on MR Phase, Magnetic Susceptibility and Electrical Properties Mapping, Santiago, Chile, 2024. *Oral*

2025

63. Multi-compartment susceptibility source separation (mc-chi-separation) using GRE data only: Integrated model of QSM and myelin water imaging
H. Shin, S. Saidha, J. Lee, P. Calabresi, P. van Zijl, X. Li. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Oral*

64. Accelerating Multi-parametric MRI Sequences Using Learned Acquisition and Reconstruction Optimization (LARO): A Retrospective Study
J. Zhang, **H. Shin**, B. Dewey, S. W. Remedios, A. Carass, X. Li, P. van Zijl, S. Saidha, P. Calabresi, J. L. Prince. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Poster*

65. Metric-based Automatic Quality Assessment for Quantitative susceptibility mapping in UK Biobank study
Z. Xu, **H. Shin**, X. Li, Y. Qiao. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Poster*

66. Quantitative susceptibility mapping, chi-separation (χ-separation), and their associations with cognitive functions in the UK Biobank study
Y. S. Hong, **H. Shin**, Z. Xu, J. Lee, J. Koo, H. Jeong, D. E. Arking, E. Guallar, P. van Zijl, Y. Qiao, X. Li. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Poster*

67. Impacts of Cerebral Iron Accumulation on Cognitive Changes and Motor Dysfunction in Aging: A Quantitative Susceptibility Mapping (QSM) Study
J. Bae, Z. Gong, A. Guo, N. Fox, N. Zhang, A. De Rouen, **H. Shin**, X. Li, M. Bouhrara. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Poster*

68. DeepSepSTI-R_2*: R_2-Free Anisotropic Susceptibility Source Separation in Susceptibility Tensor Imaging with Deep Learning
Z. Fang, **H. Shin**, B. E. Dewey, P. A. Calabresi, P. van Zijl, J. Sulam, X. Li. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Oral*

69. Development of χ-separation pipeline for UK Biobank dataset
J. Koo, H. Jeong, J. Kim, R. Hong, **H. Shin**, X. Li, Y. S. Hong, Y. Qiao, D. E. Arking, J. Lee. 33rd ISMRM Annual Meeting, Honolulu, HI, USA, 2025. *Oral*

70. MRI Assessment of Blood-Brain Barrier Disruption is Highest in the Penumbral Region of Vascular White Matter Lesions
S. Sasannia, L. Knutsson, M. Matsyuk, S. Wang, J. Wang, **H. Shin**, J. Prince, R. Leigh, P. van Zijl, P. Nyquist. International Stroke Conference 2025, Los Angeles, CA, USA, 2025 (*Stroke* 56(Suppl 1):TP184). *Poster*

71. Ten Year Increases in White Matter Hyperintensity Volume Correlates with the Severity of Ongoing Blood Brain Barrier Permeability when Measured with Dynamic Susceptibility Contrast MRI
S. Sasannia, R. Leigh, S. Wang, M. Matsyuk, J. Zhang, **H. Shin**, J. Prince, L. Becker, L. Yanek, D. Vaidya, P. van Zijl, L. Knutsson, P. Nyquist. International Stroke Conference 2025, Los Angeles, CA, USA, 2025 (*Stroke* 56(Suppl 1):WMP21). *Poster*

72. Increased Severity of Blood-Brain Barrier Disruption is Associated with Greater Burden of Cerebrovascular Disease in Asymptomatic Patients
R. Leigh, S. Sasannia, S. Wang, M. Matsyuk, J. Zhang, **H. Shin**, J. Prince, L. Becker, L. Yanek, D. Vaidya, P. van Zijl, L. Knutsson, P. Nyquist. International Stroke Conference 2025, Los Angeles, CA, USA, 2025 (*Stroke* 56(Suppl 1):DP15). *Poster*

2026

73. Iron mapping in Parkinson's disease using deep learning-enhanced susceptibility source separation
H. Shin, K. A. Mills, T. M. Dawson, T. Kim, J. Lee, X. Li, P. van Zijl, J. Prasuhn. 34th ISMRM Annual Meeting, Cape Town, South Africa, 2026. *Oral* · ***Magna Cum Laude Merit Award***

74. White matter hyperintensity burden as a predictor of white matter damage
H. Shin, S. Sasannia, M. Matsyuk, S. Wang, J. Zhang, X. Li, F. Szczepankiewicz, J. Prince, R. Leigh, L. Knutsson, P. van Zijl, P. Nyquist. 34th ISMRM Annual Meeting, Cape Town, South Africa, 2026. *Poster*

75. HiFi-QSM: A deep learning framework for high-resolution QSM reconstruction
C. Oh, T. Kim, J. Kim, **H. Shin**, J. Lee. 34th ISMRM Annual Meeting, Cape Town, South Africa, 2026. *Flash presentation*

76. Submillimeter mapping of diffusion and susceptibility MRI to characterize cortical microstructures
Y. Uchida, **H. Shin**, L. Gomez-Isaza, J. C. Troncoso, A. Bibic, J. Gerold, D. Sung, Y. Jun, F. Wang, Z. Dong, S. Huang, B. Bilgic, P. van Zijl, X. Li, K. Oishi. 34th ISMRM Annual Meeting, Cape Town, South Africa, 2026. *Flash presentation*

77. MWF-MIMOSA for Highly-efficient Joint Relaxometry and Myelin Water Fraction Mapping
Y. Chen, Y. Jun, **H. Shin**, S. Li, S. Fujita, X. Yong, J. Kim, J. Lee, G. F. Piredda, T. Hilbert, H. Liu, H. Ye, B. Gagoski, K.-S. Chan, B. Bilgic. 34th ISMRM Annual Meeting, Cape Town, South Africa, 2026. *Oral* · ***Summa Cum Laude Merit Award***